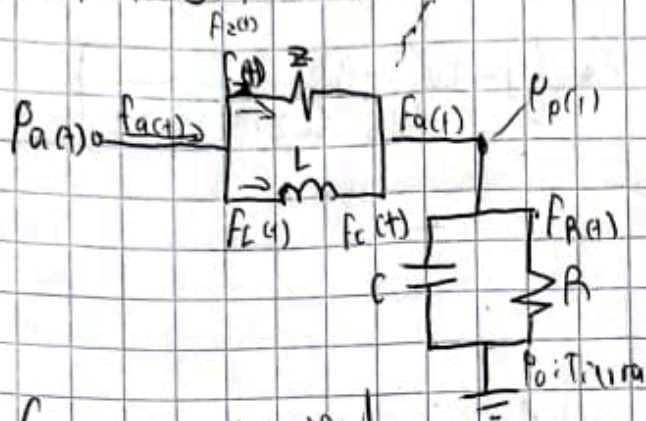


Practica 5.4: Sistema Cardiovascular



Ecuacion principal

$$F_a(t) = F_z(t) + F_L(t) = F_c(t) + F_R(t)$$

$$F_z(t) = \frac{P_a(t) - P_p(t)}{Z} \quad F_c(t) = \frac{C d P_p(t)}{dt}$$

$$F_L(t) = \frac{1}{L} \int [P_a(t) - P_p(t)] dt \quad F_R(t) = \frac{P_p(t)}{R}$$

Procedimiento algebraico

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{1}{L} \int [P_a(s) - P_p(s)] dt = \frac{C d P_p(s)}{dt} + \frac{P_p(s)}{R}$$

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{LS} = \frac{CS P_p(s) + \frac{P_p(s)}{R}}{1}$$

$$\left(\frac{1}{Z} + \frac{1}{LS}\right) P_a(s) = \left(CS + \frac{1}{R} + \frac{1}{Z} + \frac{1}{LS}\right) P_p(s)$$

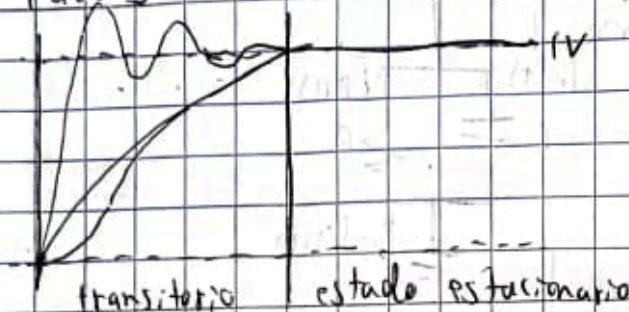
$$\frac{LS+2}{LZS} P_a(s) = \frac{CLS^2 + (CS + RLS + RZ) P_p(s)}{RLZS}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{RLZS}{CLS^2 + (CLZ + RL)S + RZ} = \frac{RLS + RZ}{CLAS^2 + (CLZ + RL)S + RZ}$$

Error ^{en estado} estacionario

$$e(s) = \lim_{s \rightarrow 0} s P_a(s) \left[1 - \frac{P_p(s)}{P_a(s)} \right] = \lim_{s \rightarrow 0} s \cdot \frac{1}{s} \left[1 - \frac{R_1 s + R_2}{CLR_2 s^2 + (LZ + RL)s + R_2} \right]$$

$$= 1 - \frac{R_2}{R_2} = 0V$$



$$\frac{R_1 s + R_2}{CLR_2 s^2 + (LZ + RL)s + R_2}$$

Estabilidad en lazo abierto

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = CLR_2 \quad ; \quad \lambda_{1,2} = \frac{-(LZ + RL) \pm \sqrt{(LZ + RL)^2 - 4(LR_2^2)}}{2CLR_2} = \frac{(-)(+)}{+} \Rightarrow \frac{R_2}{R_2}$$

$$b = LZ + RL$$

$$c = R_2$$

El sistema tiene una respuesta estable porque $\text{Re} \lambda_{1,2} < 0$

Modelo de ecuaciones integra-diferenciales

$$P_p(t) \left(\frac{Z + R}{Z} \right) = \frac{P_a(t)}{Z} + \frac{1}{Z} \int [P_a(t) - P_p(t)] dt - \left(\frac{dP_p(t)}{dt} \right) \frac{ZR}{Z + R}$$

$$P_p(t) = \frac{P_a(t)}{Z} + \frac{1}{Z} \int [P_a(t) - P_p(t)] dt - \left(\frac{dP_p(t)}{dt} \right) \frac{ZR}{Z + R}$$

